Let $n \geq 2$. Suppose $\mathcal{U} \in \mathcal{U}_0(B_n)$. (i.e. $\mathcal{U}(x,y) = I$ if $(x,y) \in E_n^0$.)

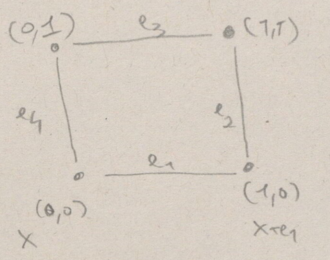
Then $\|I - \mathcal{U}(x,y)\|_{HS} \leq \left(2|x|_1 \, S_{B_n}(\mathcal{U})\right)^{1/2}$ for all $(x,y) \in E_n$.

$$\|I - \mathcal{U}(x,y)\|_{HS} = \sqrt{\mathrm{Tr}\left((I - \mathcal{U}_{xy})^*(I - \mathcal{U}_{xy})\right)}$$

$\|\cdot\| = \sqrt{\langle \cdot, \cdot \rangle}$

in other words, smallness of the Wilson action, in local regions (notice the $|x|_1$), implies smallness of the unitaries on each edge, for configurations in the axial gauge.

As a counterexample, consider $d=2$ and B_2:



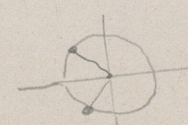
The Wilson action in this case is just $\simeq$

$\|I - \mathcal{U}(e_1)\mathcal{U}(e_2)\mathcal{U}(e_3)\mathcal{U}(e_4)\|$.

Consider $\mathcal{U}(1) = S^1 \subseteq \mathbb{C}$.

$e^{i\theta_1} e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$

$\theta_1 + \theta_2 = 0 \Rightarrow \theta_1 = -\theta_2$



for instance $\mathcal{U}(e_1) = e^{i\,3\pi/4}$

and $\mathcal{U}(e_2) = e^{i(-3\pi/4)}$.

$\mathcal{U}(e_3) = 1$, $\mathcal{U}(e_4) = 1$.

Then $\|1 - \mathcal{U}(e_1)\| \nleq 0 = \left(2|(0,0)|_1 S_{B_2}(\mathcal{U})\right)^{1/2}$.

Proof. If $x = 0$, the result is clear b/c then $(x,y) \in E_n^0$.

Note that unitaries are normal, b/c $\mathcal{U}^*\mathcal{U} = I$, $\mathcal{U}\mathcal{U}^* = I \Rightarrow \mathcal{U}\mathcal{U}^* = \mathcal{U}^*\mathcal{U}$.

Hence, you have a spectral thm for unitaries.

One can show that $\|I - \mathcal{U}(x,y)\|$ is bounded by $\sum_{p \in B(x)} \|I - \mathcal{U}_p\|$

for $B(x)$ containing $\leq |x|_1$ plaquettes. just follow the tree & use induction.

The Wilson action is $\sum_{p \in P(B_n)} \frac{1}{2}\|I - \mathcal{U}(p)\|^2$, $\cancel{\geq \sum_{p \in B(x)} \|I - \mathcal{U}_p\|^2}$

$\cancel{\geq \frac{1}{2}\|I - \mathcal{U}(x,y)\|}$